

IET, DAVV, Indore

B.E. II (IT-A & B)

Class Test II (Oct.-2023)

3AIRC1: Discrete Mathematics

Max Marks: 20

Time: 70 min

Note: Attempt any **FOUR** questions. All questions carry equal marks.

Q.1. Use Huffman coding to encode the following symbols with the frequencies listed: A: 0.08, (05)
B: 0.10, C: 0.12, D: 0.15, E: 0.20, F: 0.35. What is the average number of bits used to
encode a character? (05)

Q.2. Using mathematical induction, prove that $n! > 3^n$, for $n \geq 7$. (05)

Q.3. Solve the recurrence relation $x_n = 2x_{n-1} + 15x_{n-2}$, $x_1 = 2$, $x_2 = 4$. (05)

Q.4. Check whether the following is a tautology or a contradiction: (05)

$$[(p \vee q) \wedge (p \rightarrow r)] \rightarrow r$$

Q. State the converse, inverse and contrapositive of the following implication:

"I am awake and I do not dream of home when I do not work hard."
